



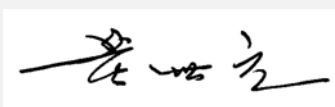
JIANGSU RUDONG LINGYANG WIND POWER PROJECT



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Summary:

China Classification Society Certification Co., Ltd. (CCSC), commissioned by Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd., has performed the third periodic verification of Jiangsu Rudong Lingyang Wind Power Project (hereafter referred to as "the project") covering the monitoring period from 01/01/2013 - 05/12/2017, on the basis of requirements of Verified Carbon Standard (VCS) Version 4.

The project is a grid connected renewable energy project located in Jiangsu Province. The objective of the project is to generate electricity using state-of-the-art wind power generation technology and to sell into the State Power Grid. The project achieves CO₂ emission reduction by replacing electricity generated by fossil fuel fired power plant connected into East China Power Grid.

Totally 33 sets of 1500kW wind turbines have been installed, providing a total capacity of 49.5MW. The area has relative rich wind resource. It is estimated that the annual generation of the project is 106.7 GWh. As a result, 98,526 tonnes of CO₂ emission reduction is expected to be generated.

The project reduces greenhouse gas emissions by supplying zero-emission wind power to East China Power Grid, which replaces the same amount of electricity generated by fossil fuel fired power plants connected into East China Power Grid, and therefore, avoids the CO₂ emissions in generating the same amount of electricity provided by the fossil fuel fired power plants.

The baseline scenario of the project is the electricity supply of equal amount as the project from the East China Power Grid. The baseline scenario of the project is the same against the scenario prior to the start of the implementation of the project activity.

The project began to construct on 10-Jun-2007, the first wind turbine was put into operation on 06-Dec-2007 and the project started full operation on 12-Jan-2008.

The project has been registered as a VCS project (VCS Project ID: 294) under VCS Standard Version 2007.1 using approved CDM methodology ACM0002 (Version 06) "Consolidated methodology for grid-connected electricity generation from renewable sources", with the first renewable crediting period from 06/12/2007 to 05/12/2017. The VCS monitoring report of this monitoring period has been reviewed against the relevant criteria and decisions by the VERRA Board, including the approved baseline and monitoring methodology ACM0002 (Version 06).

The verification process includes three phases:

- 1) Desk review of documents;
- 2) On-site visit and follow-up interviews;
- 3) Issuance of the final verification report and opinion.

CCSC appointed a qualified verification team in accordance with internal procedures to perform the verification. During the verification, 1 Clarification Request (CL), no Corrective Action Requests (CARs) and no forward request (FAR) were raised.

As a result of this verification, the verification team confirms that:

- All operations of the project are implemented and installed as planned and described in the validated Project Description;
- The registered monitoring plan is in accordance with the applied approved CDM methodology, i.e., ACM0002 (Version 06) ;
- The implementation of monitoring follows the registered monitoring plan;
- The monitoring system is in place and functional;
- The installed equipment for measuring parameters required for calculating emission reductions are calibrated appropriately.

Based on the information observed and evaluated, the verification team confirms that the emission reductions are correctly calculated in the MR (version 02 dated 18/07/2022).

In conclusion, it is CCSC's opinion that the project activity "Jiangsu Rudong Lingyang Wind Power Project", as described in the MR (version 02 dated 18/07/2022), meets all relevant requirements for VCS and all relevant host Party criteria. Hence, CCSC is able to certify that the emission reductions from the project during the monitoring period from 01/01/2013 to 05/12/2017 amount to 375,674 tCO₂e and requests issuance of the equivalent VCUs.

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1 INTRODUCTION

1.1 Objective

Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd., has commissioned CCSC to verify and provide a verification statement of the emission reductions of the Jiangsu Rudong Lingyang Wind Power Project (hereafter referred to as "the project"), which is located in Rudong County, Jiangsu Province of China for the monitoring period from 01/01/2013 to 05/12/2017, based on requirement of VCS Standard (version 4.3) /11/ as well as criteria to provide for consistent project operations, monitoring and reporting.

CCSC as the validation/verification body (VVB) of the project has been accredited as a DOE by UNFCCC and meets the competence requirements as set out in ISO 14065:2020.

Verification of emission reductions from a project activity is the independent review and ex post determination by VVB of the monitored reductions in GHG emissions during the reported monitoring period. The objectives of verification are to:

- a) Ensure that the project activity has been implemented and operated as per the registered CDM-PDD /3/ and that all physical features (technology, project equipment, and monitoring and metering equipment) of the project are in place;
- b) Ensure that the monitoring report and other supporting documents provided are complete in accordance with the latest applicable version of Registration and Issuance Process /15/ and in accordance with the additional requirements stated by the VERRA;
- c) Ensure that actual monitoring systems and procedures comply with the monitoring systems and procedures described in the monitoring plan and the approved methodology /12/;
- d) Evaluate the data recorded and stored as per the monitoring methodology /12/.

1.2 Scope and Criteria

The verification scope encompasses an independent and objective review and ex-post determination of the monitored reductions in GHG emissions by the CCSC. The verification scope covers the relevant documents (e.g. the CDM-PDD /3/, the monitoring plan, the VCS monitoring report (VCS-MR) /1/, the emission reduction calculation spreadsheet /2/, supporting documents available to the verifier and information collected through performing interviews and during the on-site assessment, VERRA's requirements publicly available, relevant rules, including the host country legislation, etc.) to be independently reviewed, the project geographical locations to be visited on-site, the related project local stakeholders to be interviewed with, and processes that are necessary to acquire objective evidence for the evaluation of the project compliance to the VCS requirements and associated interpretations.

The above verification activities are conducted according to the VERRA's requirements. In doing so, the principles of accuracy and completeness, relevance, reliability and credibility were followed.

The verification is not meant to provide any consulting service towards the PPs. However, stated requests for clarifications and/or corrective actions may provide input for improvement of the project.

1.3 Level of Assurance

CCSC has undertaken a reasonable assurance engagement in accordance with VCS Standard (version 4.3) /11/. It requires a reasonable level of assurance in verification that GHG assertions are free of material errors, omissions and misrepresentations. The verification conclusion is based on the CDM-PDD /3/, VCS-MR /1/, and supporting evidences made available to the verifier and information collected through performing interviews and during the the on-site inspection.

1.4 Summary Description of the Project

The project is a grid connected renewable energy project located in Jiangsu Province. The objective of the project is to generate electricity using state-of-the-art wind power generation technology and to sell into the State Power Grid. The project achieves CO₂ emission reduction by replacing electricity generated by fossil fuel fired power plant connected into East China Power Grid.

Totally 33 sets of 1500kW wind turbines have been installed, providing a total capacity of 49.5MW. The area has relative rich wind resource. It is estimated that the annual generation of the project is 106.7 GWh. As a result, 98,526 tonnes of CO₂ emission reduction is expected to be generated.

The project reduces greenhouse gas emissions by supplying zero-emission wind power to East China Power Grid, which replaces the same amount of electricity generated by fossil fuel fired power plants connected into East China Power Grid, and therefore, avoids the CO₂ emissions in generating the same amount of electricity provided by the fossil fuel fired power plants.

The baseline scenario of the project is the electricity supply of equal amount as the project from the East China Power Grid. The baseline scenario of the project is the same against the scenario prior to the start of the implementation of the project activity.

The project began to construct on 10-Jun-2007, the first wind turbine was put into operation on 06-Dec-2007 and the project started full operation on 12-Jan-2008.

The project was registered as a CDM project on 12/01/2009 (UNFCCC registration reference number of 2169), and registered as a VCS project as well (VCS ID 294).

During the monitoring period from 01/01/2013 to 05/12/2017, the total GHG emission reductions generated by the project are 375,674 tCO₂e.

2 VERIFICATION PROCESS

2.1 Method and Criteria

The overall verification, from Contract Review to Verification Report & Opinion, was conducted using CCSC internal procedures.

CCSC verified the project against the requirements set in VCS/11//13//14//15/, and other relevant VCS requirements.

The following sections outline each step in more detail.

2.2 Document Review

A desk review of the registered CDM-PDD /3/, the registered validation report /4/, the previous verification reports /5/, the applied monitoring methodology /12/, the VCS monitoring report (VCS-MR) /1/ and supporting documents was conducted by the verification team. The aim of the desk review of the documentation was to verify the completeness of the data and the information presented, to carry out the compliance check of the VCS-MR /1/ with respect to the monitoring plan and the applied methodology /12/. Particular attention was given to the quality of the metering equipment including calibration requirements /9//10/, the monitored data and its evidence, and the emission reduction calculation. The evaluation

of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions was also conducted.

Appendix B of this report contains a complete list of all documents and proofs reviewed by the verification team.

2.3 Interviews

On 06/07/2022, CCSC performed an on-site assessment at the physical site of the project. The verification team verified that the actual implementation of the project was in compliance with that described in the CDM-PDD /3/, VCS-MR /1/. This included the review of the project operation based on the evidence of on-site observation and presented documents.

During the on-site assessment, the verification team has interviewed with key personnel from the project owner (the project proponent) and the consultancy.

The assessment content and topics/the persons interviewed at the on-site assessment were provided in the following table.

Date: 06/07/2022	
Interview topics	Interviewed Organizations and persons
--The status of VCS project implementation. -- Any changes of the VCS project. -- The project on-site inspection -- The evidences of construction status and operation of key equipment, parameters monitoring and data processing activities, monitor equipment and calibration. -- Monitoring data. -- Quality Management; organizational structure, responsibilities and competencies; Internal QA/QC Management procedures and document control. -- Compliance with National Laws and Regulations.	Jiangsu Longyuan Wind Power Co., Ltd. Yang Jianjun, Director of the plant Chen Zhibao, Operation Staff
-- The impact of the project activity. -- The complaint by local stakeholders. -- The stakeholder consultation during the operation of the project activity.	Yu Lijun, Rudong Ecology and Environment Protection Bureau Xu Meilin, local resident of Lingyang reclamation area
-- Preparation of Monitoring Report. -- Compliance of the monitoring plan with the monitoring methodology. -- Compliance of monitoring with the monitoring plan	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd., Mr. Xue Qiang, Project Manager

Date: 06/07/2022	
Interview topics	Interviewed Organizations and persons
-- Assessment of data and calculation of GHG emission reductions.	

2.4 Site Inspections

The on-site assessment involved following activities,

An assessment of the implementation and operation of the project activity as per the CDM-PDD /3/;

A review of information flows for generating, aggregating and reporting the monitoring parameters;

Interviews with relevant personnel to confirm that the operational and data collection procedures are implemented in accordance with the approved monitoring plan;

A cross-check between information provided in the monitoring report /1/ and data from other sources such as operational records or similar data sources;

A check of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the CDM-PDD /3/ and the selected methodology /12/;

A review of calculations and assumptions made in determining the GHG data and emission reductions;

An identification of quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

2.5 Resolution of Findings

During the verification of a project activity, to identify issues that need to be further elaborated upon, researched or added to in order to confirm that the project activity meets the VERRA's requirements and can achieve credible emission reductions, the verification team will issue a Corrective Action Request (CAR), a Clarification Request (CL) or a Forward Action Request (FAR) depending on different situations.

The objective of this phase of the verification is to resolve issues related to the monitoring, implementation and operations of the registered project activity that could impair the capacity of the registered project activity to achieve emission reductions or influence the monitoring and reporting of emission reductions prior to CCSC's positive conclusion on the GHG emission reduction calculation.

Findings established during the verification can either be seen as a non-fulfilment of criteria ensuring the proper implementation of a project or where a risk to deliver high quality emission reductions is identified.

A Corrective Action Request (CAR) will be raised if one of the following occurs:

- a) Non-compliance with the monitoring plan or methodology are found in monitoring and reporting and has not been sufficiently documented by the project proponents, or if the evidence provided to prove conformity is insufficient;
- b) Modifications to the implementation, operation and monitoring of the registered project activity has not been sufficiently documented by the project proponents;
- c) Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions;
- d) Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the project proponents.

A Clarification Request (CL) will be raised if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

A Forward Action Request (FAR) will be raised, for actions if the monitoring and reporting require attention and/or adjustment for the next verification period.

1 Clarification Request (CL) has been raised during the current verification.

2.6 Forward Action Requests

No Forward Action Requests were raised during the current verification.

2.7 Eligibility for Validation Activities

The VVB of "China Classification Society Certification Co., Ltd. " commissioned by the project proponent is one of eligible VVBs listed on the VERRA website (<https://verra.org/project/vcs-program/validation-verification/>), and it holds the accreditation for validation/verification for the sectoral scope: "1. Energy (renewable/non-renewable)", to which the project activity belongs. Therefore, CCSC is eligible for the validation/verification of the project activity.

3 VALIDATION FINDINGS

3.1 Participation under Other GHG Programs

The project was validated by TUEV-SUED on 12/01/2009 under CDM scheme for registration /4/. CCSC only performed verification activities on the proposed monitoring period for the project according to the VCS Program Guide.

3.2 Methodology Deviations

No deviation from the methodology is needed to be submitted.

3.3 Project Description Deviations

As described in Annex 1 of MR version 01 dated 6-May-2009/5/, only emission reductions achieved from 06/12/2007 to 11/01/2009 will be considered as VCUs. Thus the crediting period was from 06/12/2007 to 11/01/2009.

A deviation is requested for the project crediting period. The first crediting period has been changed from 06/12/2007 - 11/01/2009 to 06/12/2007-05/12/2017 as the deviation.

As per Appendix 3 of VCS standard version 4.3, Registered projects and projects that complete validation on or before 19 March 2020 remain eligible to apply the crediting period requirements under VCS standard version 3. The project is registered under VCS Standard Version 2007.1 and completed validation before 19/03/2020. Thus the crediting period requirements under VCS standard version 3 has been applied by the project.

As per section 3.8 of VCS Standard version 3. The project crediting period shall be a maximum of ten years and may be renewed at most twice for non-AFOLU projects. Projects registered under other GHG programs are not eligible for VCU issuance beyond the end of the total project crediting period under those programs. Renew validation report shall be issued after the end of the (previous) project crediting period but within two years after the end of the (previous) project crediting period. The project started operation on 06/12/2007. The end date of the CDM 3*7 crediting period of the project is 11/01/2030. Thus the total VCS crediting period was determined from 06/12/2007 to 11/01/2030, of which the first VCS crediting period is from 06/12/2007~05/12/2017. Since the project did not conduct the validation of crediting period renewal within the two years after the end of the first VCS crediting period, the VCS crediting period of the project ends on 05/12/2017.

According to the information collected during on-site visit and the relevant evidence provided by project owner, CCSC can confirm that the deviation does not impact the applicability of the methodology, additionality or the appropriateness of the baseline scenario, and the project remains in compliance with the applied methodology, the deviation has been appropriately described and justified in the monitoring report and will be described in all subsequent monitoring reports, and the project remains in compliance with the VCS Program rules.

3.4 Grouped Project

Not applicable as the project activity is not grouped project.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

4.1.1 Project Implementation in accordance with the CDM-PDD.

The project has been registered as VCS project activity with the Project ID 294 (<https://registry.verra.org/app/projectDetail/VCS/294>).

The project is a grid connected renewable energy project located in Jiangsu Province. The objective of the project is to generate electricity using state-of-the-art wind power generation technology and to sell into the State Power Grid. The project achieves CO₂ emission reduction by replacing electricity generated by fossil fuel fired power plant connected into East China Power Grid.

Totally 33 sets of 1500kW wind turbines have been installed, providing a total capacity of 49.5MW. The area has relative rich wind resource. It is estimated that the annual generation of the project is 106.7 GWh. As a result, 98,526 tonnes of CO₂ emission reduction is expected to be generated.

The project reduces greenhouse gas emissions by supplying zero-emission wind power to East China Power Grid, which replaces the same amount of electricity generated by fossil fuel fired power plants connected into East China Power Grid, and therefore, avoids the CO₂ emissions in generating the same amount of electricity provided by the fossil fuel fired power plants.

The baseline scenario of the project is the electricity supply of equal amount as the project from the East China Power Grid. The baseline scenario of the project is the same against the scenario prior to the start of the implementation of the project activity.

The project began to construct on 10-Jun-2007, the first wind turbine was put into operation on 06-Dec-2007 and the project started full operation on 12-Jan-2008.

Through the on-site visit, the power generating technology, installed capacity, technical specifications of wind turbine and generator, and power delivering systems indicated in the CDM-PDD are in place and the project has been operated as per the CDM-PDD /3/. The verification team confirms the project has been implemented as per the registered CDM-PDD /3/. During this monitoring period, the monitoring of this project activity was in compliance with the registered CDM-PDD.

The major technical specifications of the project have been verified during the site visit and listed in the following table and the verification team confirmed they are consistent with the registered CDM-PDD /3/.

Table 4-1 The main technical parameters of wind turbines

Operating data	
Rated capacity:	1500kW
Cut-in wind speed:	4m/s
Cut-out wind speed (10 min. avg.):	25m/s
Rated wind speed:	13m/s
Wind Class - IEC:	Ib
Rotor	
Number of rotor blades:	3
Rotor diameter:	70.5m
Swept area:	3904m ²
Rotor speed(variable):	12.0-22.2rpm
Tower	
Hub heights-IEC:	54.7/64.7m
Power control	Active blade pitch control
Gearbox	Three step planetary spur gear system
Generator	Doubly fed, three-phase induction (asynchronous)
Converter	Pulse-width modulated IGBT frequency converter
Braking system (fail-safe)	Electromechanical pitch control for each blade (3 self-contained systems)
	Hydraulic parking brake
Yaw system	Electromechanical driven with wind direction sensor and automatic cable unwind
Control system	PLC (Programmable logic controller) with remote control and monitoring system
Noise reduction	Impact noise insulation of the gearbox and generator
	Sound reduced gearbox
	Noise reduced nacelle
	Rotor blades with minimized noise level
Lightning protection system	Lightning receptors installed along blades
	Surge protection in electrical components
Tower design	Multi-coated, conical tubular steel tower with safety ladder to the nacelle
	Load lifting system, load-bearing capacity over 200 kg

According to the registered CDM-PDD /3/ and the emission reductions calculation spreadsheet /2/, quantity of net electricity generation supplied by the project plant to the grid is 406,839.41 MWh during this monitoring period. All the data and the calculated process in the emission reductions calculation spreadsheet /2/ have been checked by the verification team and it is correctly and in accordance with both the monthly reading records/7/ and Electricity Transaction Notes (ETNs)/8/. Therefore, the verification team confirmed there are no changes of the project from the registered CDM PDD /3/ that cause increase of emission reductions.

By checking UNFCCC website (<https://cdm.unfccc.int/Projects/DB/TUEV-SUED1218638332.96/view>), it is confirmed by CCSC that the project has been registered as CDM project activity on 12/01/2009 with reference No. 2169. GHG emission reductions have been issued as CERs from 12/01/2009 - 31/12/2012 under CDM scheme. There are no CDM issuance during this monitoring period. Furthermore, CCSC checked public

information from the REC Mechanism database of China, IREC Mechanism database, JI database, Chinese Emission Trading System, Gold Standard Registry and interviewed with project owner during site visit, it is confirmed that except CDM and VCS scheme, the project has not been participated or been rejected under any other GHG programs since validation or previous verification.

Furthermore, as per “Notice on Strengthening Enterprise Greenhouse Gas Emission Report Management” issued by Ministry of Ecology and Environment of P.R.China, China has a national emissions trading scheme only cover the high-emission industries, including thermal power generation, petrochemical, chemical, building materials, iron and steel, non-ferrous, paper, aviation and other key emission industries that emitted at least 26,000 tons of CO₂e/year. By checking “List of key emissions in the national carbon emissions transaction quota management in 2019-2020” issued by Ministry of Ecology and Environment of P.R.China, the project owner is not included in the mandatory emission control scheme and thus no emission cap was enforced for the project activity.

Therefore, the verification team confirmed that the project only applies CERs and VCU under CDM and VCS, and no rejection from CDM and VCS occurs, there are no other forms of environmental credits applied or issued for the project activity, the emission reduction resulted from the project during this monitoring period would only apply for VCUs.

By checking China’s National Plan on Implementation of the 2030 Agenda for Sustainable Development/16/ and 17 SDGs defined by UNDP, and interviewing with stakeholders during site visit, CCSC confirmed that the project would contribute to sustainable development as follow.

- **SDG 13:** Reducing greenhouse gas emissions compared to a business-as-usual scenario. During this monitoring period, the emission reductions achieved were 375,674tCO₂e. By checking UNFCCC website (<https://cdm.unfccc.int/Projects/DB/DNV-CUK1262853821.64/view>) and VERRA website (<https://registry.verra.org/app/projectDetail/VCS/294>), it is confirmed that the cumulative Contributions Over Project Lifetime were 851,664tCO₂e achieved emission reductions since the project started operation.

- **SDG 7:** Providing clean and renewable energy source and displacing the power generation of fossil fuel power plants, reducing pollution emissions caused by coal burning significantly, thus mitigating the air pollution and its adverse impacts on human health, promoting sustainable economic development in local area. During this monitoring period, the renewable energy supplied to East China Power Grid was 406,839.41MWh. By checking UNFCCC website (<https://cdm.unfccc.int/Projects/DB/TUEV-SUED1218638332.96/view>) and VERRA website (<https://registry.verra.org/app/projectDetail/VCS/294>), it is confirmed that the cumulative Contributions Over Project Lifetime was 922,316.879MWh renewable energy supplied to East China Power Grid since the project started operation.

- **SDG 8:** Providing direct and indirect employment opportunities during construction and

operation period, which promotes sustained, inclusive and sustainable economic growth, full and productive employment and decent work for local residents. By checking labor contract and interview, it is confirmed that since operation of the project activity, 6 long term working opportunities are provided.

As per Instructions for Completing the Monitoring Report (Version 4.1), the cumulative impact should be calculated by summing the current project contributions with all impacts included since the project start date. Hence, **CL-1** was raised requesting PP to update Table 1 of the MR accordingly. In the updated MR, cumulative SDG impacts of the project activity has been updated which is in line with Instructions for Completing the Monitoring Report (Version 4.1). Therefore, **CL-1** was closed.

In conclusion, according to the information collected during site visit and the relevant evidence provided by project owner, CCSC can confirm that:

- a) No project design change occurred during this monitoring period and the project was implemented in accordance the registered CDM-PDD /3/;
- b) The project contributes to sustainable development including: creating job opportunities in the project area, displacing part of the electricity generated by coal-fired power plants, and generating emission reductions;
- c) The GHG emission reductions or removals generated by the project have not included in an emissions trading program or any other mechanism that includes GHG allowance trading;
- d) The project has not sought or received another form of GHG-related environmental credit, including renewable energy certificates, during this monitoring period. None of the credits will be claimed in the same period;
- e) The GHG emission reductions generated by the project during this monitoring period (01/01/2013~05/12/2017) will be only claimed under VCS program as VCUs for the project to avoid double counting.

4.1.2 Compliance of monitoring with the monitoring plan

The verification team has assessed the implementation of monitoring by the project owner against the registered monitoring plan. A short summary on the verification of every parameter listed in the monitoring plan and used for emission reduction calculation is provided as follow.

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Data / Parameter: (as in the monitoring plan of PD):	EG _y Net electricity delivered to the grid by the project (MWh)

Measuring frequency:	Continuously measurement
Reporting frequency:	Monthly recorded
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes
Type of monitoring equipment:	Calculated via equation: $EG_y = EG_{\text{export},y} - EG_{\text{import},y}$
Is accuracy of the monitoring equipment as stated in the PD? If the PD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practise?	Yes. For details see Table 4-2.
Calibration frequency /interval:	Annually as per relevant sectoral standards in China
Is the calibration interval in line with the monitoring plan of the PD? If the PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practise?	Yes
Company performing the calibration:	See Table 4-2
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	Yes
Is(are) calibration(s) valid for the whole reporting period?	Yes. For details see Table 4-2.
If applicable, has the reported data been cross-checked with other available data?	Yes, the reported data has been cross-checked with Electricity Transaction Notes in line with registered CDM-PDD /3/.
How were the values in the monitoring report verified?	The verification details have been reported in Section 4.4 of this report.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of	Yes.

emission reductions and are necessary QA/QC processes in place?	
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied or has a request for deviation been approved?	Not applicable.

Data / Parameter: (as in the monitoring plan of PD):	EG _{export,y} Electricity generation supplied to the power grid by the project in year y (MWh)
Measuring frequency:	Continuously measurement
Reporting frequency:	Monthly recorded
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes
Type of monitoring equipment:	Electricity meters
Is accuracy of the monitoring equipment as stated in the PD? If the PD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practise?	Yes. For details see Table 4-2.
Calibration frequency /interval:	Annually as per relevant sectoral standards in China
Is the calibration interval in line with the monitoring plan of the PD? If the PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practise?	Yes
Company performing the calibration:	See Table 4-2
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	Yes

Is(are) calibration(s) valid for the whole reporting period?	Yes. For details see Table 4-2.
If applicable, has the reported data been cross-checked with other available data?	Yes, the reported data has been cross-checked with Electricity Transaction Notes which is in line with description in the registered CDM-PDD /3/.
How were the values in the monitoring report verified?	The verification details have been reported in Section 4.4 of this report.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied or has a request for deviation been approved?	Not applicable.

Data / Parameter: (as in the monitoring plan of PD):	$EG_{import,y}$ Electricity imported from the power grid by the project in year y (MWh)
Measuring frequency:	Continuously measurement
Reporting frequency:	Monthly recorded
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes
Type of monitoring equipment:	Electricity meters
Is accuracy of the monitoring equipment as stated in the PD? If the PD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practise?	Yes. For details see Table 4-2.

Calibration frequency /interval:	Annually as per relevant sectoral standards in China
Is the calibration interval in line with the monitoring plan of the PD? If the PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practise?	Yes
Company performing the calibration:	See Table 4-2
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	Yes
Is(are) calibration(s) valid for the whole reporting period?	Yes. For details see Table 4-2.
If applicable, has the reported data been cross-checked with other available data?	Yes, the reported data has been cross-checked with Electricity Transaction Notes which is in line with description in the registered CDM-PDD /3/.
How were the values in the monitoring report verified?	The verification details have been reported in Section 4.4 of this report.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied or has a request for deviation been approved?	Not applicable.

According to the applied methodology ACM0002 (Version 06) , Net electricity delivered to the grid by the project (EG_y) is calculated as the difference of electricity generation supplied to the power grid by the project ($EG_{export,y}$) and electricity imported from the power grid by the project ($EG_{import,y}$), which is used for the calculation of baseline emission.

Furthermore, the verification team has verified that the accuracy of measure of the monitoring equipment was set according to the requirements and all calibration procedures were carried out according to the registered CDM-PDD and manufacturer specifications. The calibration information of the monitoring meters /10/ in this monitoring period has been listed in Table 4-2. The monitoring equipment has been installed in the project in line with the registered CDM-PDD. 2 meters (key meter M1 and check meter M2) are installed in the project. The electricity supplied to the grid and from the grid are both measured according to the key meter M1. The check meter M2 is for backup usage when the key meter is out of order. During this monitoring period, both the meters worked well.

In summary, the verification team is able to verify that the accuracy the monitoring equipment was set according to the registered CDM-PDD /3/ and relevant sectoral standard of China.

Table 4-2 Verified calibration information of monitoring meters

Meter	Type	Serial number	Accuracy	Frequency	Calibration date	Valid until	Calibration Entity
M1	bidirectional electricity meter	85093984	0.2S	Annually	18- Apr -2012	17-Apr -2013	Jiangsu Provincial Institute of Metrology
					10-Mar-2013	09-Mar-2014	
					10-Feb-2014	09-Feb-2015	
					10-Jan-2015	09-Jan-2016	
					10-Dec-2015	09-Dec-2016	
					10-Nov-2016	09-Nov-2017	
					10-Oct-2017	09-Oct-2018	

M2	bidirectional electricity meter	85060035	0.2S	Annually	18- Apr -2012	17-Apr -2013	
					10-Mar-2013	09-Mar-2014	
					10-Feb-2014	09-Feb-2015	
					10-Jan-2015	09-Jan-2016	
					10-Dec-2015	09-Dec-2016	
					10-Nov-2016	09-Nov-2017	
					10-Oct-2017	09-Oct-2018	

The verification team confirmed that there is no proposed or actual change to monitoring plan of registered CDM-PDD during this monitoring period. All required equipments and procedures are available and implemented in an appropriate manner. All necessary monitoring instruments are installed. All required instruments including standby and operating procedures for the same have been implemented in an appropriate manner. The project is completely operational and the same has been confirmed on-site. Neither mistakes nor malfunction on meters have been observed during this monitoring period.

4.2 Safeguards

4.2.1 No Net Harm

By checking the EIA summary and conclusion provided in the registered CDM-PDD, it is confirmed that wind power is green power and the impact caused by wind power on the surrounding ecosystem and residents, wastewater, solid waste and atmosphere etc. is very little, there would be no net harm caused due to the project activity. Besides, the EIA of the project are approved by the government.

Also, no potential environment or social economic matter was found during on-site assessment. The project is renewable energy project and thus no net harm observed in air or water quality on-site.

4.2.2 Local Stakeholder Consultation

As per the registered CDM-PDD, On 21-Nov-2006, under the support of local government, the project owner successfully held a stakeholder meeting in Rudong County. Totally 14 stakeholder representatives participated the meeting, respectively from the Development and Reform Bureau of Rudong County, the Environmental Protection Bureau of Rudong County, the Management Council of Rudong Dayanggang Development Zone, and the Sanmin Village of Rudong County. The survey showed that the local people supported and agreed with the construction of the project. They believe that the project would have a positive impact on the local economic development and their living conditions, without having negative impacts on the environment and the ecosystem.

Communications with Local stakeholders was being carried out at periodic intervals. There are no negative comments received for the project.

All such conclusion has been verified through on-site assessment and check registered CDM-PDD.

4.3 AFOLU-Specific Safeguards

The project is not AFOLU project, and thus this section is not applicable for the project.

4.4 Accuracy of GHG Emission Reduction and Removal Calculations

4.4.1 Parameters and information flow

The parameters required by the monitoring plan and the way CCSC has verified the information flow (from data generation, aggregation, to recording, calculation and reporting for these parameters including the values) in the monitoring report are described as follow.

Parameters monitored ex-post:

$$EG_y = EG_{\text{export},y} - EG_{\text{import},y}$$

Where:

- 1) **EG_y : Net electricity delivered to the grid by the project (MWh);**
- 2) **$EG_{\text{export},y}$ is electricity generation supplied to the power grid by the project in year y (MWh);**
- 3) **$EG_{\text{import},y}$ is electricity imported from the power grid by the project in year y (MWh);**

According to the VCS-MR /1/ of the project and the registered CDM-PDD /3/, the electricity generation supplied to the power grid by the project in year y ($EG_{\text{export},y}$) and electricity imported from the power grid by the project in year y ($EG_{\text{import},y}$) are both continuously monitored through the meter which is both agreed by the grid company and the project owner. The Net electricity delivered to the grid by the project is the difference of electricity generation supplied to the power grid by the project in year y and electricity imported from the power grid by the project in year y, i.e. $EG_y = EG_{\text{export},y} - EG_{\text{import},y}$. The daily and monthly monitoring records of the project have been verified during on-site assessment.

The verification team has cross-checked EG_y , $EG_{\text{export},y}$, and $EG_{\text{import},y}$ in the emission reductions calculation spreadsheet /2/ with Electricity Transaction Notes /8/ of the project from the grid company. By the on-site assessment, the values in emission reductions calculation spreadsheet /2/ is in line with Electricity Transaction Notes /8/ during this period.

Parameter determined ex-ante:

EF_y : The combined baseline emission factor of the East China Power Grid.

According to the registered CDM-PDD /3/, the EF_y is ex-ante determined as 0.9234 tCO₂/MWh for East China Power Grid over the first crediting period. The emission factors used in this monitoring report have been verified against the registered CDM-PDD /3/ and found them to be consistent.

4.4.2 Assessment Data and Calculation

It was verified that the project is a newly built wind power plant. Therefore, the project emission of the project is zero under ACM0002 (Version 06) /12/ ($PE_y=0$).

The project is a new wind power plant. Therefore, no leakage needs to be considered for the project activity according to the methodology /12/. ($LE_y=0$).

Since the project emission and the leakage emission has been justified as zero under ACM0002 (Version 06) /12/, the calculation of the emission reductions simply involves the calculation of the baseline emissions.

Formula 1:

$$ER_y = BE_y - 0 - 0 = EG_y \times EF_y \quad (1)$$

Where:

ER_y : Emission reductions in year y (tCO₂/yr)

BE_y : The baseline emissions (tCO₂/yr)

EG_y : Net electricity delivered to the grid by the project (MWh)

EF_y : The combined baseline emission factor of the East China Power Grid (tCO₂/MWh)

Net electricity delivered to the grid by the project (EG_y) is calculated as the difference of electricity generation supplied to the power grid by the project in year y ($EG_{\text{export},y}$) and electricity imported from the power grid by the project in year y ($EG_{\text{import},y}$), as follow:

$$\text{Formula 2: } EG_y = EG_{\text{export},y} - EG_{\text{import},y} \quad (2)$$

By checking Electricity Transaction Notes /8/ during this period, it is confirmed that the transaction date of $EG_{\text{export},y}$ and $EG_{\text{import},y}$ is 24:00 of last day of each month. By checking Confirmation Letter local power grid company /23/, it is confirmed that the electricity data for the period of 01/01/2013-30/11/2017 and 01/12/2017-05/12/2017 are sourced from daily meter records and has confirmed by the local power grid company.

Following the approach described in section 4.4.1 above, $EG_{\text{export},y}$, $EG_{\text{import},y}$ and EG_y were verified as follow.

Table 4-3 The verified $EG_{\text{export},y}$, $EG_{\text{import},y}$ and EG_y of the project (Unit: MWh)

Period	Electricity exported to the grid by the project ($EG_{\text{export},y}$) (MWh)			Electricity imported from the grid by the project ($EG_{\text{import},y}$) (MWh)			Net electricity supplied to the grid EG_y (MWh)
	Values from main meter (M1) readings	Values from ETNs	Conservative	Values from main meter (M1) readings	Values from ETNs	Conservative	G=C-F
	A	B	C=Min (A, B)	D	E	F=Max (D, E)	
01-Jan-2013 to 31-Jan-2013	5,900.88	5,900.88	5,900.88	38.88	38.88	38.88	5,862.00
01-Feb-2013 to 28-Feb-2013	7,507.04	7,507.04	7,507.04	30.69	30.69	30.69	7,476.35
01-Mar-2013 to 31-Mar-2013	12,193.13	12,193.13	12,193.13	17.30	17.30	17.30	12,175.83
01-Apr-2013 to 30-Apr-2013	11,276.74	11,276.74	11,276.74	9.56	9.56	9.56	11,267.18
01-May-2013 to 31-May-2013	8,954.28	8,954.28	8,954.28	14.12	14.12	14.12	8,940.16
01-Jun-2013 to 30-Jun-2013	6,523.57	6,523.57	6,523.57	21.48	21.48	21.48	6,502.09
01-Jul-2013 to 31-Jul-2013	7,108.84	7,108.84	7,108.84	15.03	15.03	15.03	7,093.81
01-Aug-2013 to 31-Aug-2013	7,375.90	7,375.90	7,375.90	17.85	17.85	17.85	7,358.05
01-Sep-2013 to 30-Sep-2013	5,611.44	5,611.44	5,611.44	29.12	29.12	29.12	5,582.32
01-Oct-2013 to 31-Oct-2013	9,512.67	9,512.67	9,512.67	11.96	11.96	11.96	9,500.71
01-Nov-2013 to 30-Nov-2013	6,756.61	6,756.61	6,756.61	19.28	19.28	19.28	6,737.33
01-Dec-2013 to 31-Dec-2013	6,791.22	6,791.22	6,791.22	25.81	25.81	25.81	6,765.41
01-Jan-2013 to 31-Dec-2013	95,512.32	95,512.32	95,512.32	251.08	251.08	251.08	95,261.24
01-Jan-2014 to 31-Jan-2014	7,640.10	7,640.10	7,640.10	32.00	32.00	32.00	7,608.10
01-Feb-2014 to 28-Feb-2014	9,909.85	9,909.85	9,909.85	27.46	27.46	27.46	9,882.39
01-Mar-2014 to 31-Mar-2014	10,121.31	10,121.31	10,121.31	11.96	11.96	11.96	10,109.35
01-Apr-2014 to 30-Apr-2014	7,487.52	7,487.52	7,487.52	30.81	30.81	30.81	7,456.71
01-May-2014 to 31-May-2014	8,628.80	8,628.80	8,628.80	17.80	17.80	17.80	8,611.00
01-Jun-2014 to 30-Jun-2014	4,780.40	4,780.40	4,780.40	33.58	33.58	33.58	4,746.82

01-Jul-2014 to 31-Jul-2014	4,661.29	4,661.29	4,661.29	30.28	30.28	30.28	4,631.01
01-Aug-2014 to 31-Aug-2014	5,890.42	5,890.42	5,890.42	52.92	52.92	52.92	5,837.50
01-Sep-2014 to 30-Sep-2014	8,849.05	8,849.05	8,849.05	20.16	20.16	20.16	8,828.89
01-Oct-2014 to 31-Oct-2014	10,205.46	10,205.46	10,205.46	26.34	26.34	26.34	10,179.12
01-Nov-2014 to 30-Nov-2014	6,045.40	6,045.40	6,045.40	31.54	31.54	31.54	6,013.86
01-Dec-2014 to 31-Dec-2014	7,430.97	7,430.97	7,430.97	29.47	29.47	29.47	7,401.50
01-Jan-2014 to 31-Dec-2014	91,650.57	91,650.57	91,650.57	344.32	344.32	344.32	91,306.25
01-Jan-2015 to 31-Jan-2015	7,536.19	7,536.19	7,536.19	24.43	24.43	24.43	7,511.76
01-Feb-2015 to 28-Feb-2015	7,333.73	7,333.73	7,333.73	34.92	34.92	34.92	7,298.81
01-Mar-2015 to 31-Mar-2015	4,077.18	4,077.18	4,077.18	25.82	25.82	25.82	4,051.36
01-Apr-2015 to 30-Apr-2015	9,343.69	9,343.69	9,343.69	24.60	24.60	24.60	9,319.09
01-May-2015 to 31-May-2015	6,175.43	6,175.43	6,175.43	20.49	20.49	20.49	6,154.94
01-Jun-2015 to 30-Jun-2015	5,520.69	5,520.69	5,520.69	22.96	22.96	22.96	5,497.73
01-Jul-2015 to 31-Jul-2015	5,328.20	5,328.20	5,328.20	35.79	35.79	35.79	5,292.41
01-Aug-2015 to 31-Aug-2015	6,138.17	6,138.17	6,138.17	43.72	43.72	43.72	6,094.45
01-Sep-2015 to 30-Sep-2015	4,394.18	4,394.18	4,394.18	40.85	40.85	40.85	4,353.33
01-Oct-2015 to 31-Oct-2015	4,496.52	4,496.52	4,496.52	39.14	39.14	39.14	4,457.38
01-Nov-2015 to 30-Nov-2015	7,026.08	7,026.08	7,026.08	35.02	35.02	35.02	6,991.06
01-Dec-2015 to 31-Dec-2015	4,845.18	4,845.18	4,845.18	42.65	42.65	42.65	4,802.53
01-Jan-2015 to 31-Dec-2015	72,215.24	72,215.24	72,215.24	390.39	390.39	390.39	71,824.85
01-Jan-2016 to 31-Jan-2016	8,078.18	8,078.18	8,078.18	27.80	27.80	27.80	8,050.38
01-Feb-2016 to 29-Feb-2016	6,985.51	6,985.51	6,985.51	29.10	29.10	29.10	6,956.41
01-Mar-2016 to 31-Mar-2016	8,034.98	8,034.98	8,034.98	27.96	27.96	27.96	8,007.02
01-Apr-2016 to 30-Apr-2016	6,319.43	6,319.43	6,319.43	27.34	27.34	27.34	6,292.09
01-May-2016 to 31-May-2016	7,557.79	7,557.79	7,557.79	27.93	27.93	27.93	7,529.86
01-Jun-2016 to 30-Jun-2016	4,002.89	4,002.89	4,002.89	28.25	28.25	28.25	3,974.64

01-Jul-2016 to 31-Jul-2016	4,059.01	4,059.01	4,059.01	29.12	29.12	29.12	4,029.89
01-Aug-2016 to 31-Aug-2016	4,569.88	4,569.88	4,569.88	24.64	24.64	24.64	4,545.24
01-Sep-2016 to 30-Sep-2016	7,444.31	7,444.31	7,444.31	39.79	39.79	39.79	7,404.52
01-Oct-2016 to 31-Oct-2016	8,284.56	8,284.56	8,284.56	12.04	12.04	12.04	8,272.52
01-Nov-2016 to 30-Nov-2016	7,537.41	7,537.41	7,537.41	30.68	30.68	30.68	7,506.73
01-Dec-2016 to 31-Dec-2016	7,080.53	7,080.53	7,080.53	21.23	21.23	21.23	7,059.30
01-Jan-2016 to 31-Dec-2016	79,954.48	79,954.48	79,954.48	325.88	325.88	325.88	79,628.60
01-Jan-2017 to 31-Jan-2017	5,109.41	5,109.41	5,109.41	29.78	29.78	29.78	5,079.63
01-Feb-2017 to 28-Feb-2017	7,187.82	7,187.82	7,187.82	24.14	24.14	24.14	7,163.68
01-Mar-2017 to 31-Mar-2017	6,556.34	6,556.34	6,556.34	23.61	23.61	23.61	6,532.73
01-Apr-2017 to 30-Apr-2017	7,546.16	7,546.16	7,546.16	15.85	15.85	15.85	7,530.31
01-May-2017 to 31-May-2017	6,448.73	6,448.73	6,448.73	20.73	20.73	20.73	6,428.00
01-Jun-2017 to 30-Jun-2017	4,141.94	4,141.94	4,141.94	28.50	28.50	28.50	4,113.44
01-Jul-2017 to 31-Jul-2017	3,178.34	3,178.34	3,178.34	29.69	29.69	29.69	3,148.65
01-Aug-2017 to 31-Aug-2017	5,700.84	5,700.84	5,700.84	43.76	43.76	43.76	5,657.08
01-Sep-2017 to 30-Sep-2017	5,769.24	5,769.24	5,769.24	30.32	30.32	30.32	5,738.92
01-Oct-2017 to 31-Oct-2017	10,621.16	10,621.16	10,621.16	19.95	19.95	19.95	10,601.21
01-Nov-2017 to 30-Nov-2017	6,763.27	6,763.27	6,763.27	30.97	30.97	30.97	6,732.30
01-Dec-2017 to 05-Dec-2017	92.91	92.91	92.91	0.39	0.39	0.39	92.52
01-Jan-2017 to 05-Dec-2017	69,116.16	69,116.16	69,116.16	297.69	297.69	297.69	68,818.47
Total	408,448.77	408,448.77	408,448.77	1,609.36	1,609.36	1,609.36	406,839.41

The verification team has confirmed the emission reductions of the project have been conservatively calculated in line with above mentioned Formula 1&2 in the emission reduction calculation spreadsheet /2/. The verification team has checked and confirmed the emission reduction is to be correctly accounted. The total emission reduction is as follow:

Table 4-5 Verified Emission Reductions during this monitoring period

Period	EG _y (MWh)	EF _y (tCO ₂ /MWh)	BE _y (tCO ₂ e)	PE _y (tCO ₂ e)	LE _y (tCO ₂ e)	ER _y (tCO ₂ e)
	A	B	C=A*B	D	E	F=C-D-E
01-Jan-2013 to 31-Dec-2013	95,261.24	0.9234	87,964	0	0	87,964
01-Jan-2014 to 31-Dec-2014	91,306.25	0.9234	84,312	0	0	84,312
01-Jan-2015 to 31-Dec-2015	71,824.85	0.9234	66,323	0	0	66,323
01-Jan-2016 to 31-Dec-2016	79,628.60	0.9234	73,529	0	0	73,529
01-Jan-2017 to 05-Dec-2017	68,818.47	0.9234	63,546	0	0	63,546
Total	406,839.41	/	375,674	0	0	375,674

4.4.3 Comparison of ERs

Based on the above assessment, CCSC can conclude that the emission reduction during the monitoring period from 01/01/2013 to 05/12/2017 have been quantified correctly in accordance with the project description and applied methodology /12/, and the emission reduction during this monitoring period is verified as 375,674 tCO₂e. Compared with the estimated emission reductions under the same days of 1800 from 01/01/2013 to 05/12/2017 in the registered CDM-PDD /3/, 485,881 (equals to 103,412*1800/365) tCO₂e, the verified emission reduction is 22.68% less than the estimated value. CCSC considers the value of verified emission reduction is conservative.

4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

The evidences used to determine the GHG emission reductions are assessed as follows:

Assessment of the Evidence Quality	
Parameters monitored	EG _y EG _{export,y} EG _{import,y}
Monitoring method	Net electricity delivered to the grid by the project (EG _y) is calculated as the difference of electricity generation supplied to the power grid by the project in year y (EG _{export,y}) and electricity imported from the

	power grid by the project in year y ($EG_{import,y}$). $EG_{export,y}$ and $EG_{import,y}$ are measured by key meter M1 and check meter M2.
Measuring/Reading/Recording frequency	The parameters $EG_{export,y}$ and $EG_{import,y}$ are continuously measured and monthly recorded.
Calibration frequency/ interval	See Table 4-2 Verified calibration information of monitoring meters. Through on-site assessment and checking the calibration records of meters, the team confirmed that all the monitoring meters meet the rated accuracy level and the calibration frequency fulfils the requirement as described in the monitoring plan.
Calibration cover the monitoring period	The verification team confirms that the validity period of all conducted calibrations covers this monitoring period from 01/01/2013 to 05/12/2017.
Data reported	Through cross checking with Electricity Transaction Notes /8/, the monitored data can be confirmed as acceptable to calculate emission reductions.

In conclusion, all necessary documentation is collected, referenced and is easily accessible in hard-copy or electronic format. The data pertaining to the monitored parameters are maintained in the identified internal records and consistent with the values stated in the Monitoring Report /1/ and ER calculation spreadsheet /2/. Monitoring data have been cross-checked with Electricity Transaction Notes.

4.6 Non-Permanence Risk Analysis

The project is not AFOLU project, and thus non-permanence risk analysis is not applicable for the project.

5 VERIFICATION CONCLUSION

CCSC has carried out the verification of the project "Jiangsu Rudong Lingyang Wind Power Project". The verification was performed based on the requirements set by Voluntary Carbon Standard (VCS) version 4.3 /11/, as well as criteria given to provide for consistent project operations, monitoring and reporting. This verification covers the period from 01/01/2013 to 05/12/2017 inclusive.

In the course of the verification 1 Clarification Request (CL), no Corrective Action Requests (CARs) and no forward request (FAR) was raised. The verification is based on the VCS-MR /1/, the CDM-PDD /3/, ER calculation spread sheet /2/ and supporting documents available to CCSC.

As the result of the verification, CCSC confirms that:

- The project has been implemented and operated as per the registered CDM-PDD /3/ and that all physical features (technology, project equipment, and monitoring and metering equipment) of the project are in place;
- The VCS-MR /1/ and other supporting documents provided are complete in accordance with the latest applicable version of the VCS Registration & Issuance Process /15/ and in accordance with the additional requirements stated by the VERRA;
- The actual monitoring systems and procedures comply with the monitoring systems and procedures described in the registered CDM-PDD and the approved methodology i.e. ACM0002 (Version 06) /12/;
- The GHG emission reductions are calculated without material misstatements and in a conservative and appropriate manner.

Based on the evidence and information that are considered necessary to guarantee that GHG emission reductions are appropriately calculated, CCSC confirms the following statement:

Verification period: From 01/01/2013 to 05/12/2017

Verified GHG emission reductions and removals in the above verification period:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
01-Jan-2013 to 31-Dec-2013	87,964	0	0	87,964
01-Jan-2014 to 31-Dec-2014	84,312	0	0	84,312
01-Jan-2015 to 31-Dec-2015	66,323	0	0	66,323
01-Jan-2016 to 31-Dec-2016	73,529	0	0	73,529
01-Jan-2017 to 05-Dec-2017	63,546	0	0	63,546
01-Jan-2013 to 05-Dec-2017	375,674	0	0	375,674

5 VERIFICATION CONCLUSION

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As the result of the verification, CCSC confirms that:

- The project has been implemented and operated as per the registered CDM-PDD /3/ and that all physical features (technology, project equipment, and monitoring and metering equipment) of the project are in place;
- The VCS-MR /1/ and other supporting documents provided are complete in accordance with the latest applicable version of the VCS Registration & Issuance Process /15/ and in accordance with the additional requirements stated by the VERRA;
- The actual monitoring systems and procedures comply with the monitoring systems and procedures described in the registered CDM-PDD and the approved methodology i.e. ACM0002 (Version 06) /12/;
- The GHG emission reductions are calculated without material misstatements and in a conservative and appropriate manner.

Based on the evidence and information that are considered necessary to guarantee that GHG emission reductions are appropriately calculated, CCSC confirms the following statement:

Verification period: From 01/01/2013 to 05/12/2017

Verified GHG emission reductions and removals in the above verification period:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)

01/01/2013~05/12/2017	375,674	0	0	375,674
Total	375,674	0	0	375,674

APPENDIX A: ABBREVIATIONS

CAR	Corrective Action Request
CDM	Clean Development Mechanism
CER	Certified Emission Reduction(s)
CL	Clarification request
CO ₂	Carbon Dioxide
CO ₂ e	Carbon Dioxide Equivalent
CCSC	China Classification Society Certification Co., Ltd.
DOE	Designated Operational Entity
DNA	Designated National Authority
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reduction
ETN	Electricity Transaction Note
FAR	Forward Action Request
GHG	Greenhouse Gas(es)
MR	Monitoring Report
ECPG	East China Power Grid
PD	Project Description
PDD	Project Design Document
PP	Project Proponent
UNFCCC	United Nations Framework Convention on Climate Change
VVB	Validation/Verification Body
VCS	Verified Carbon Standard
VCU	Verified Carbon Unit

APPENDIX B: REFERENCE

Documents provided by the Project proponent and relevant background documents have been reviewed or referenced for the periodic verification conclusions.

Ref no.	Reference Document
/1/	Monitoring report version 01 dated 18/05/2022, version 02 date 18/07/2022
/2/	ER calculation spreadsheet
/3/	Registered CDM PDD
/4/	Registered Validation Report
/5/	Previous Verification Report and corresponding MR
/6/	Project Proponent Registration Representation
/7/	Monthly reading records
/8/	Electricity Transaction Notes (ETNs) issued by the grid company for this monitoring period
/9/	Qualification certificate of the meter calibration entity
/10/	Calibration reports of the meters
/11/	VERRA, VCS Standard, version 4.3
/12/	CDM Executive Board, ACM0002 (Version 06) Consolidated methodology for grid-connected electricity generation from renewable sources
/13/	VERRA, Monitoring Report Template Form version 4.1
/14/	VERRA, VCS Program Guide version 4.2
/15/	VERRA, Registration & Issuance Process version 4.2
/16/	China's National Plan on Implementation of the 2030 Agenda for Sustainable Development

/17/	Chinese DNA, Notification on Determining Baseline Emission Factor of China's Grid issued on 02/07/2009
/18/	China National Economic Trading Commission, Technical Administrative Code of Electric Energy Metering, DL/T 448-2016
/19/	The State Bureau of Quality and Technical Supervision, Technical Norm of the Calibration of AC Watthour Meters at place of installation, JJG 1055-1997
/20/	Business License
/21/	Power purchase agreement
/22/	Electrical primary wiring diagram
/23/	Confirmation Letter local power grid company
/24/	Questionnaire survey

APPENDIX C: RESOLUTION OF CLARIFICATION REQUESTS AND CORRECTIVE ACTION REQUESTS AND FORWARD ACTION REQUESTS

Table 1. CL from this verification

CL ID	CL-1	Section no.	4.1.1	Date: 13/07/2022
Description of CL				
As per Instructions for Completing the Monitoring Report (Version 4.1), the cumulative impact should be calculated by summing the current project contributions with all impacts included in previously approved VCS monitoring reports or Sustainable Development Contribution Reports. Please update Table 1 of the MR accordingly.				
Project proponent response				Date: 18/07/2022
The calculation of cumulative impact of the project has been revised in the updated MR, please refer to the Table 1 in Section 1.11.				
Documentation provided by project proponent				
updated MR				
VVB assessment				Date: 18/07/2022
In the updated MR, cumulative SDG impacts of the project activity has been updated which is in line with Instructions for Completing the Monitoring Report (Version 4.1). Therefore, CL-1 was closed.				

Table 2. CAR from this verification

CAR ID	N/A	Section no.	N/A	Date: N/A
Description of CAR				
N/A				
Project proponent response				Date: N/A
N/A				

Documentation provided by project proponent	
N/A	
VVB assessment	Date: N/A
N/A	

Table 3. FAR from this verification

FAR ID	N/A	Section No.	N/A	Date: N/A
Description of FAR				
N/A				
Project proponent response				Date: N/A
N/A				
Documentation provided by project proponent				
N/A				
VVB assessment				Date: N/A
N/A				